

Meraf Hailesslassie

Minneapolis, MN · merafhailesslassie@gmail.com · 507-581-6918

EDUCATION

St. Olaf College — Northfield, MN

Feb 2021 – Dec 2024

B.A., Mathematics; Quantitative Economics; Data Science and Statistics

- Relevant coursework: Real Analysis I, Probability Theory, Modern Computational Math, Econometrics: Time Series and Forecasting, Statistical Modeling, Introduction to Data Science

RESEARCH EXPERIENCE

Spatial Statistics Research Assistant

Jun 2023 – Dec 2023

St. Olaf College, Statistics Department — Northfield, MN

- Adapted and synthesized spatial statistical methods from multiple prior studies to estimate total volume change of Storglaciären, an internationally studied glacier in northern Sweden.
- Estimated and propagated error through Digital Elevation Models (DEMs) to quantify accuracy, precision, and uncertainty in volume-change calculations.
- Co-authored a research paper and delivered a presentation on the spatial statistical analysis of glacier volume loss, as part of St. Olaf's Collaborative Undergraduate Research and Inquiry (CURI) program.

Course Research Projects

St. Olaf College

Healthcare Access & Disparities Analysis — with Wonu Falae

- Modeled the relationship between ethnicity, gender, and access to healthcare among Medicaid beneficiaries using multiple linear regression, logistic regression, stepwise selection, and lasso/ridge regression with cross-validation.
- Identified health status, program enrollment, and utilization as the strongest predictors of access; reported a null finding for ethnicity and gender with a rigorous discussion of the dataset's limitations.

Global Air Quality & Health Analysis — with Wonu Falae

- Built an interactive R Shiny dashboard analyzing global air pollution, mortality, population, and literacy data (2010–2017) from WHO, World Bank, and Kaggle sources.
- Produced geographic visualizations of air quality patterns and their relationship to pollution-related mortality across world regions.

Handwritten Digit Classification — Team Project

- Built and compared LDA, QDA, logistic regression, multinomial regression, and cross-validated KNN models to classify handwritten digits, using custom feature engineering and hyperparameter tuning.
- Diagnosed model performance tradeoffs through confusion matrix analysis as classification complexity increased.

Research Assistant, Healthcare Economics

Spring 2023

St. Olaf College

- Developed an inpatient and outpatient Specialty Intensity Score to support evaluation of care coordination across providers.
- Processed Limited Data Set (LDS) Medicare claims data in R and identified key variables to construct a provider-diagnosis network for care-coordination analysis.

Research, Time Series and Forecasting

Fall 2023

St. Olaf College

- Investigated drivers of rising U.S. healthcare costs, including high prices, overtesting, and revenue-focused incentives.
- Built a structural VAR (vector autoregression) model to quantify the relationship between personal consumption factors and healthcare spending trends.

Research, Behavioral Economics

Spring 2024

St. Olaf College

- Led a research project applying game theory and behavioral economic principles to design food-waste reduction strategies for cafeteria sustainability initiatives.

PROFESSIONAL EXPERIENCE

Business Intelligence Report Developer

May 2025 – Present

American Dairy Queen Corporation — Bloomington, MN

- Streamline audit-combination processes using Power BI, Databricks, and Excel VBA to unify and optimize data workflows across finance, food safety, marketing, and operations datasets.
- Automate previously manual reporting processes, improving accuracy, efficiency, and productivity.
- Proactively learn and adopt new software tools as needed, integrating multiple platforms to streamline and automate reporting workflows.
- Participate in the Inclusive Culture Council to help sustain and advance DEI initiatives across the organization.

Data Science Intern

Jan 2025 – May 2025

Darcy Solutions Inc — Minneapolis, MN

- Designed and implemented an alert system to evaluate heat exchanger efficiency across multiple sites.
- Built a scalable Power Apps solution to automate tracking of well-building projects, from initial client request — such as an invoice or consultation inquiry — through completion.

Global Data Insight and Analytics Intern

Jun 2024 – Aug 2024

Ford Motor Company — Dearborn, MI

- Built and modified scripts to process supplier performance data, informing operational strategy across plants.
- Applied feature engineering techniques to forecast and mitigate premium freight events, reducing costs and improving supply chain resilience.

Labor Productivity and Benchmarking Intern

Jun 2022 – Aug 2022

Allina Health — Minneapolis, MN

- Analyzed quarterly productivity reports using Vizient and Workday to help managers benchmark hospital productivity.
- Supported implementation and training for PowerLabor, a system-wide productivity measurement application.

SKILLS

Statistical & Quantitative Methods: Spatial statistics, time series / VAR modeling, regression & logistic regression, LDA/QDA, KNN, lasso/ridge regression, cross-validation, error propagation, game theory, feature engineering

Technical: R, Python, SQL, Power BI, Databricks, Alteryx, QlikSense, Excel (VBA), Power Apps, SharePoint, R Shiny

Languages: English (fluent), Amharic (fluent)